Faculty of Computing – University of Sri Jayewardenepura

Object-Oriented Programming – Assignment 1

“Slide Puzzle” is a fun game that consists of numbered square-shaped movable tiles with a single space (with the exact size of a tile) to the entire board of the puzzle. The puzzle can be solved by moving the tiles horizontally or vertically to the empty space. Initially, the numbers are in random order, and the goal is to arrange the tiles as their numbers in the correct order. The following figure shows the initial board of the puzzle, valid movements as well as the winning state of the puzzle.

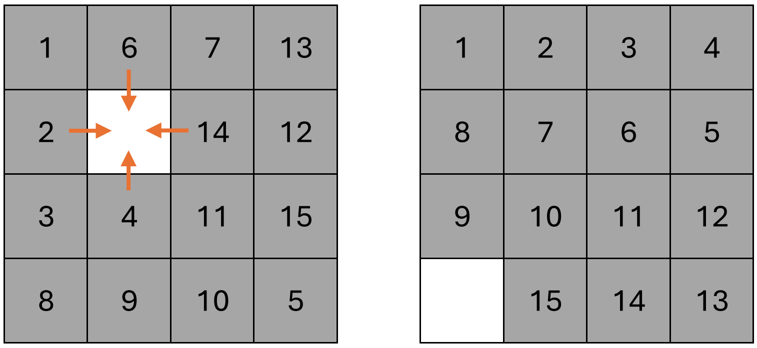

Figure 1: **Left**: Initial state of the puzzle with valid movements shown with the arrows, **Right**: the solution of the puzzle, after a series of movements.

You are asked to develop the above program using Java Swing UI framework and implement the following given features.

- The program should contain a graphical user interface (GUI) that facilitates all valid interactions with the puzzle.
- The GUI should provide visual feedback to the user regarding all the valid interactions as well as the status changes of the puzzle such as “the user wins”.
- The initial placement of the tiles should be changed whenever a user starts the puzzle.
- Your program should be able to recognize the winning state and notify the user on the winning event.
- You should give two optional constraints namely time countdown, and maximum number of moves.
    - If a user would like to opt into any of the above constraints, he or she can play till the constraint(s) are violated (i.e.. Time elapse/achieved maximum number of movements)
- The puzzle should show the best timing so far with the player’s name and should be updated as the other users break that record.
- You may use other media elements such as images and sounds as they improve the attractiveness of the game.

Bonus Marks:
4x4 Slide Puzzle can be solved by a computer program in practical time. You can earn up to 50% bonus marks (which will be added to your Exam and Assignment mark total and cap by 100%) by implementing an automated puzzle solver with the following features.

- The program should be able to give the steps of solving the puzzle starting from the current state of the board.
- The correct set of steps should be graphically visualized by moving the tiles step by step automatically if the user prompts the program to automatic solution.

Submission:
Archive the following files into a single file, rename it to your index number, and upload it to the given portal in Google Classroom.

1. Java project with the source code.
2. Screenshots of your program (at main events of the GUI)
3. JAR file build (standalone build of the program).